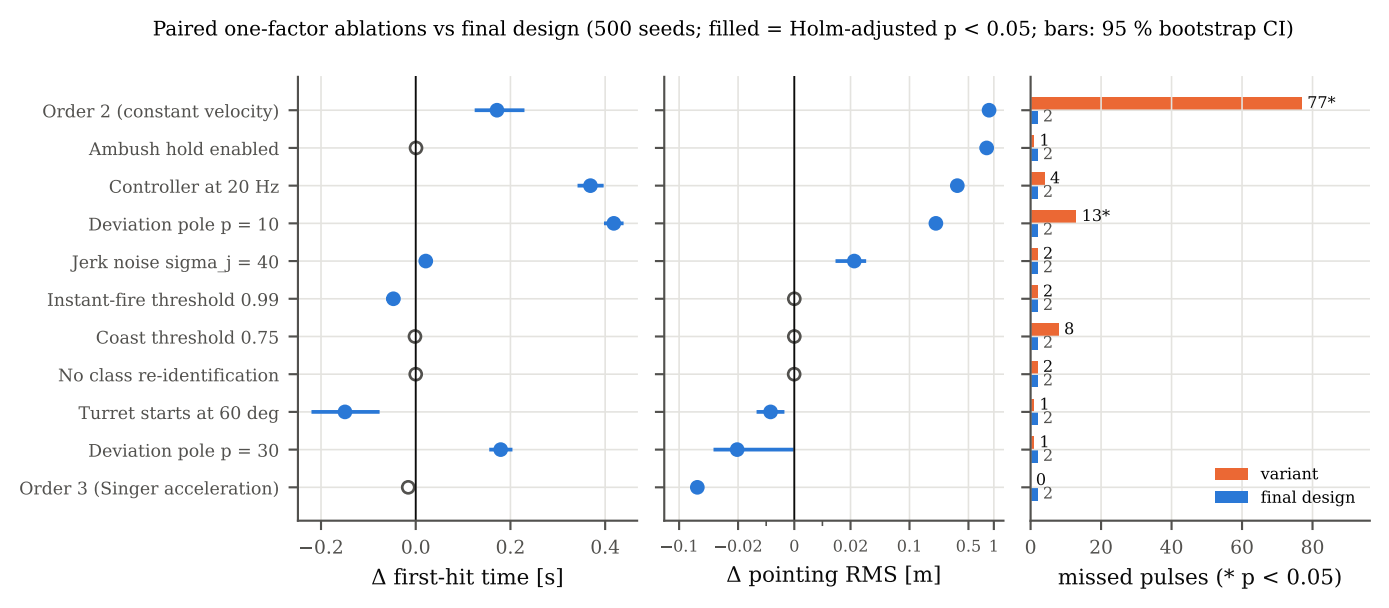
Paired one-factor ablations vs final design (500 seeds; filled = Holm-adjusted p < 0.05; bars: 95 % bootstrap CI)
Order 2 (constant velocity)
Ambush hold enabled
Controller at 20 Hz
Deviation pole p = 10
Jerk noise sigma_j = 40
Instant-fire threshold 0.99
Coast threshold 0.75
No class re-identification
Turret starts at 60 deg
Deviation pole p = 30
Order 3 (Singer acceleration)
−0.2
0.0
0.2
0.4
Δ first-hit time [s]
−0.1
−0.02
0
0.02
0.1
0.5
1
Δ pointing RMS [m]
77*
2
1
2
4
2
13*
2
2
2
2
2
8
2
2
2
1
2
1
2
0
2
variant
final design
0
20
40
60
80
missed pulses (* p < 0.05)